Semester Project : Exploring network summary plots

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December 29, 2022

1 Introduction

Network systems are an appropriate way of representing artificial and natural structures in our day-to-day lives. For instance, network-form structures could go from the different relations in social media to the whole nervous system in eukaryotes. As an effect, in order to understand those structures, it is necessary to convert them as a network system. However, computers have a limit in handling data that are too big, and understanding the shape of it by having a glance on the visualisation of the network is not obvious. The article [1] shows a proper way of extracting important aspects of a network system and to have an overview of the structures inside. Based on its topological properties, the paper suggests a way to represent the network in form of a violin plot, then applied the method to real data to show the utility of it.

Contents

2 Context

2.1 Properties of a graph

A graph is a set of nodes connected to each other by edges. The edges can be directed or undirected, which means that the edges are connecting the two nodes symmetrically or asymmetrically. In addition, the nodes might contain self-loops, which means that there exists an edge that connects the node to itself. We can say that two nodes i and j are connected with a probability $\mathbb{P}(e_{i,j} = 1)$, and that the random variable $e_{i,j}$ would follow a Bernouilli distribution with a certain probability $p_{i,j}$. Nodes will then have a degree, which is the number of edges that it has, or the number of nodes that it is connected to. An adjacency matrix is the matrix containing the information of all edges of the graph, for instance $e_{i,j}$ would be at row i and column j of such matrix. Adjacency matrices can contain all the information of a graph with numerical nodes and unique weights, and it is easier to consider this form for later computation (See Section 3). Network systems can be represented as a form of graphs, and graphs have a certain mathematical properties which enable the study of its shape. In this report, the studied networks will be in the form of undirected, unweighted graphs without self-loops.

2.2 Generating models

Graphs can be generated with certain rules to follow. Some would act on the edge probabilities (such as Erdős–Rényi model or Block model), while others would have restrictions on the global shape of the graph (such as Preferential attachment model or Watts-Strogatz model).

2.2.1 Erdős–Rényi model

Graphs following the Erdős–Rényi model are such that the generation of edges between two vertices follow a Bernouilli distribution with a certain probability p. In other words, the probability that two chosen vertices i and j are connected ($\mathbb{P}(e_{i,j} = 1)$) is p.

2.2.2 Block model

The block model is a model with an existence of "communities" inside the network. Starting from a standard Erdős–Rényi graph, the probabilities of edge connection between two nodes in the same community differ from the one of connection between nodes that are in two different communities. Provided that usually a stronger connection within the communities is expected, this model results in having several clusters of nodes and only some of the edges existing between the clusters.

2.2.3 Graphon

Finite Graphs can be expressed as exchangeable arrays by the Aldous-Hoover theorem [2] [3], which means that the distribution of the random variable ξ_i is the same as the distribution of the random variables $\xi_{\pi(i)}$ where $\pi(i)$ is a permutation of i a number from 1 to n the total number of nodes. In other words, we can rearrange the rows and columns of the adjacency matrix of the graph. We can extend the exchangeability to infinite graphs (graphs having an infinite number of nodes, or taking the limit of the graph) by using the de Finetti's representation theorem [4] : it assures that the sequence is exchangeable if the data is generated by a source that makes exchangeable sequences. As a result, even if the graph is infinite, the graph is exchangeable when the mechanism of generating that data follow a particular rule such that it produces finite exchangeable arrays.

To consider the case with infinite graphs, we introduce the notion of graphon [5], which a symmetric Lebesgue measurable function defined from $[0, 1]^2$ to $[0, 1]$. Let W be a graphon of a graph G. Then if for each node we take a random variable ξ_i for i from 1 to n the total number of node following an uniform distribution in $(0, 1)$, the graph G is generated in the following way :

$$\mathbb{P}(e_{i,j} = 1) = W(\xi_i, \xi_j).$$

We can also define the norm of the space of graphons. The cut-norm of the graphon space is

$$\delta_{W,U} = \inf_{\phi,\psi} \sup_{S,T \subseteq [0,1]} \left| \int_{S \times T} W^{\phi}(x, y) - U^{\psi}(x, y) dxdy \right|,$$

where ϕ and ψ are the preserving transformations of $[0, 1]$. A graphon space with a cut-norm would be compact.

Unfortunately, the use of graphons is not always obvious. For sparse graphs, the limit of the graph would tend to an empty, so it would ignore the properties of the graph. To solve this issue, the notion of scaled graphon is introduced. It is possible to normalise the probability by taking into account the sparsity of the graph. Rather fixing the probability of the Bernouilli distribution for the edge formation to $W(\xi_i, \xi_j)$, the probability becomes $\rho_n f(\xi_i, \xi_j)$ where ρ_n is the edge density of the graph and f is a function from $[0, 1]^2$ to (R). The edge density is the ratio of existing edges over the total number of edges that the graph can have. f has the property that $\iint_{[0,1]^2} f(x, y)dxdy = 1$ because of identifiability (this is a restriction so that if the parameters are different, the distribution differs as well, and vice versa). Then, the expected value of $\mathbb{P}(e_{i,j})$ is equal to ρ_n.

2.2.4 Preference attachment model

Graphs following the preference attachment model, also called Barabási–Albert model, is a graph where the probability that a connection is established from a vertex i to another vertex j gets higher with respect to the current degree of the vertex j.[6] The network then follow a power law distribution; the phenomenon of 'rich gets richer' (the nodes with higher degree gets more chance of having more neighboring nodes) can be observed while generating the graph.

2.2.5 Watts-Strogatz model

The Watts-Strogatz model, or the 'small world' model is generated in such way that all the nodes are connected to each other with a shorter path length.[7] In this model, the nodes are considered as being on a "lattice", and the graph is first generated as all the node have a constant number of neighbors, equally partitioned to both side of the lattice. Then with a rewiring probability, each edge of a node i can be replaced by another edge that connects the node i to another node that has not been connected chosen uniformly at random, and we repeat this procedure to all the nodes of the graph. Doing this reduces the average path length drastically from about $N/2K$ to lnN/lnK as the rewiring probability gets closer to unity, where N is the total number of nodes and K is the number of neighbors set at the beginning.

2.2.6 Bipartite graph

A bipartite graph is a graph containing two different communities where the nodes do not have edges to the nodes of the same community. In other words, the only existing edges are traced between the two communities. It is possible to consider this type of graph as a graph following the block model, with zero edge probability within the communities and a probability greater than zero between the communities.

3 Method

3.1 Dominating closed walks

Although graphs can be easily represented via a form of schemes with nodes and lines (edges), or as an adjacency matrix, it is hard to trace back and see what type of distribution the edges are following just by looking at those representations. Therefore we have to consider a way of representing the graph's topology. A method to understand the graph's structure is to scan the possible closed paths of the nodes. Then starting from a local scale, it is possible to extend this information to a global scale to have an overview of the structure.

However, the possible closed walk paths that one can draw can raise drastically as the size of the graph get bigger. Therefore, we need to define paths that are essential in the understanding of the structure. A closed path has two unique features: one is the scale, in other words the number of nodes the path passes by, and the other one is the shape. As counting all types of closed paths is too costly, we prove that the ones with the simplest shapes will predominate, thus essential enough to be considered in the representations.

Among all shapes of graphs, trees and cycles have the simplest topology. Trees have the minimal distinct edges to be crossed, while cycles have the minimum number of edges per scale. Although some paths can be in a complex form, it is possible to extend and simplify the walk. Instead of passing through the nodes that have been already stepped on, the path could consider to pass through another node, which converges to the simplest shape of walks. Thus simplified walks predominate all the other walks.

Let $\mathscr{W}_k$ be a space containing all the graphs induced by closed walks of length k. Also let W_k be the set of unlabeled subgraphs induced by $\mathscr{W}_k$. Let $F', F \in W_k$, and F' an extension of F, then the following holds:

1. F'_w crosses one more vertex than F_w while the number of step does not change.

2. F'_w has no more than one more edge than F_w

3. F'_w has the same list of vertex in the same order as F_w except replacing one vertex on the path to the newly added vertex.

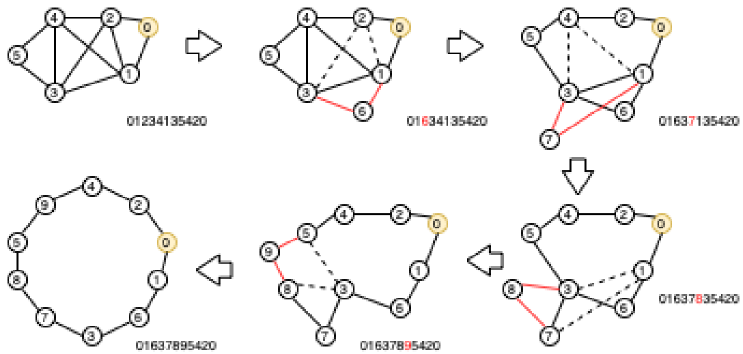

Figure 1: The scheme of the process of extending the closed walk. Each arrow represent one extension step.

Figure 1 shows the process of extending a closed walk until getting a fully extended cycle. As shown on the graph, each process involves of adding a new vertex and replacing to one of the steps, and also replacing one or two edges by two other connecting the new vertex to the other path vertices.

Let $\mathbb{E}\#\{copies\ of\ w\ in\ G\}$ be the average number of cases where the path has the same shape as the walk w. As the number of nodes of G gets larger, the following equation holds:

$$\begin{aligned} log\mathbb{E}\#\{copies\ of\ w\ in\ G\} \sim\ & const_w \\ & + \#\{nodes\ visited\ by\ w\} \cdot log\#\{nodes\ in\ G\} \\ & - \#\{edges\ traversed\ by\ w\} \cdot |log\mathbb{P}\{edges\ in\ G\}|. \end{aligned} \tag{1}$$

Equation 1 shows that the number of copies of w is influenced by the trade off between the number of nodes and the number of edges. Therefore the walk having the biggest number of copies would have the least amount of edges with the most nodes visited, which means that the fully extended closed walks will predominate all the other type of closed walks.

This domination can be extended from local to global point of view of the graph. Let $\mathscr{M}$ be the network generating mechanism. The following holds universally on the whole graph G:

$$\begin{aligned} log\mathbb{E}\#\{copies\ of\ w\ in\ G\} \sim\ & const_{w,\mathscr{M}} \\ & + \#\{nodes\ visited\ by\ w\} \cdot log\#\{nodes\ in\ G\} \\ & - |log\mathbb{P}_{\mathscr{M}}\#\{local\ copy\ of\ w\ in\ G\}|. \end{aligned} \tag{2}$$

In Equation 2, $\mathbb{P}_{\mathscr{M}}\#\{local\ copy\ of\ w\ in\ G\}$ is defined by the average number of copies of w in a local subgraph $G[u]$ over the number of copies of w in the complete version (meaning all the edges between all the nodes are present) of $G[u]$. By fixing the number of nodes that the walk would step by, the whole number of copies varies as a function of the probabilities of having copies of w in the local subgraphs. The dominance of fully extended graphs assumes the following theorem admitting the limit as the number of nodes of the graph G goes to infinity:

There exists w' an extended walk of w such that under a generating mechanism $\mathscr{M}$,

$$\#\{nodes\ in\ G\} \cdot \frac{\mathbb{P}_{\mathscr{M}}\#\{local\ copy\ of\ w'\ in\ G\}}{\mathbb{P}_{\mathscr{M}}\#\{local\ copy\ of\ w\ in\ G\}} \to \infty. \tag{3}$$

The limit 3 shows that the probability of getting a copy of w' over the one of w converge to zero slower than $1/n$ where n is the number of nodes of G. This shows that trees and cycles that are fully extended closed paths dominate over all walks universally. As a result, the average number of k-walks in G would mean counting the number of k-cycles and the number of trees which leads to the use of the algorithms on the section 3.2.

3.2 Algorithms

The algorithms were coded in Python, and the library used is `igraph`. There are two algorithms in total, which are closely linked to each other. One is about generating the violin plot by computing the prevalence of each type of tree/cycles, while the other is to determine the size of the subsamples in order to have a meaningful result out of the first algorithm. As the second algorithm uses the first one several times, it costs more in time. Therefore for larger graphs with a number of nodes greater than 10,000 will be computed with an arbitrary subsample size without using the second algorithm.

The source codes and all the elements related to the implementations are accessible in the github repository https://github.com/seorimpark/NetworkSummaryPlots.

3.2.1 Network summary plot

The first algorithm consists of drawing the network summary plot of the graph. This algorithm requires several inputs:

- The graph G: the initial graph.
- N_s: The list of sizes of the subgraphs to sample, it can be obtained by using the second algorithm presented on the section 3.2.2.
- k_{max}: the maximum scale of the cycle to consider. It must be smaller than the sizes of subgraphs to consider.
- α: the probability to reject one violin plot for one type of tree/cycle out of the final plots. It impacts the number of subsamples taken into account.

By choosing the number, the sizes of the subgraphs to sample and the maximum scale of the cycle to consider, Algorithm 1 draws a violin plot by gathering all the information from all the subsamples independently. The prevalence is computed in the following way:

- Trees: the ratio between the number of cases where three different nodes have two edges over the number of cases where three different nodes have only one edge. The first number of cases is equal to: $\#\{case\ 1\ in\ G_{|u_r|}\} = \frac{1}{2}\sum_{i=1}^{|u_r|} deg(i)(deg(i)-1)$ and the second number of cases is equal to: $\#\{case\ 2\ in\ G_{|u_r|}\} = \frac{n-2}{2}\sum_{i=1}^{|u_r|} deg(i)$ where i is a node in the list of nodes of the subsample $|u_r|$, $deg(i)$ is the degree of i and n is the length of $|u_r|$.
- Cycles: the ratio between the number of k-cycles of the subgraph out of the number of k-cycles of a complete graph having the same size of the subgraph. The number of k-cycles of the subgraph is computed by taking only the case of fully extended case out of the total closed walk of size k[8]. The k-cycle of a complete graph is computed with the following closed form formula: if we call n the size of the complete graph K_n, $\#\{k-cycle\ in\ K_n\} = \frac{n!}{2k(n-k)!}$.

For a scheme of the algorithm 1, see A.

3.2.2 Sizes of the sampled subgraphs

The second algorithm computes the adequate sizes of the subsample to take to get the final network summary plot from algorithm 1. The adequate size means a big enough size of subgraphs in order to spot prevalence of the different trees and k-cycles. The algorithm 2 takes the following input:

- The graph G: the initial graph of n nodes.
- k_{max}: the maximum scale of the cycle to consider. It must be smaller than $n/2$.
- δ: the size increment of the algorithm
- N: the number of different size the user want to get by the end
- α: the probability to reject one violin plot for one type of tree/cycle out of the final plots.

The algorithm 2 then iterates over a certain number of loops defined with δ, tests the algorithm 1 with a guess of the size of the subsample and checks if some occurrences of trees or cycles from the output t_k of algorithm 1 are equal to 0 for all the subsamples. Then we perform a test statistics on the result, and check if all the obtained p-values are below the Bonferroni correction level $\frac{\alpha}{k_{max}-1}$. If it satisfied the latter condition, or if the size is greater than $0.8n$, we take N equispaced values between 0.9 and 1.1 times the size that was chosen and return the list of final subsample sizes. Otherwise, the loop iterates back with a $1+\delta$ times bigger subsample size, and repeats the procedure considering only the problematic types of trees/cycles.

The problem of algorithm 2 is the fact that it is iterating provided that the graph has indeed occurrences of all the considering tree/cycles. As an effect, an infinite iteration has been observed with some type of graphs. For instance, graphs following the preferential attachment model with one outgoing edge have only trees, and the bipartite graphs have only trees and even-numbered cycles. Therefore, the occurrences of some type of cycles have to be zero for those graphs, which contradicts the assumption of algorithm 2. (See Section 4.1.4 and 4.1.7).

4 Application

For the network summary plots, both algorithms were used only when the number of nodes did not exceed 3000. For larger graphs, only algorithm 1 was used with a reasonable fixed size because algorithm 2 required too much computation time. For larger graphs where even algorithm 1 would be computationally heavy, another option is to use the parallelised version of the algorithm 1, which is to compute the values of t_k of one subgraph for each node simultaneously.

4.1 Generated models

Before testing the algorithms with real data, we first plotted the graphs of generated models, with certain regimes to make a particular shape of graphs. The types of model that have been used are : the Erdős–Rényi model, the preference attachment model, the Watts-Strogatz model, the block model, the model with the use of graphons, the triadic closure graph and the case of bipartite graphs.

4.1.1 Erdős–Rényi model

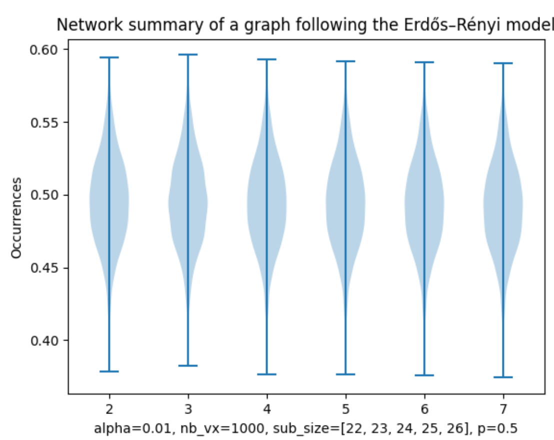

Figure 2: The network summary of a Erdős–Rényi model graph of edge probability 0.5.

Figure 2 is the network summary plot of a Erdős–Rényi model graph. The characteristic of the plots for this particular model is that the occurrences are at the same level for any type of trees or cycles. On the figure, the occurrences seem to be at around 0.5, and indeed the occurrences are similar to the edge probability which is 0.5. This property holds for other edge probabilities (See Appendix B.1)

4.1.2 Block model

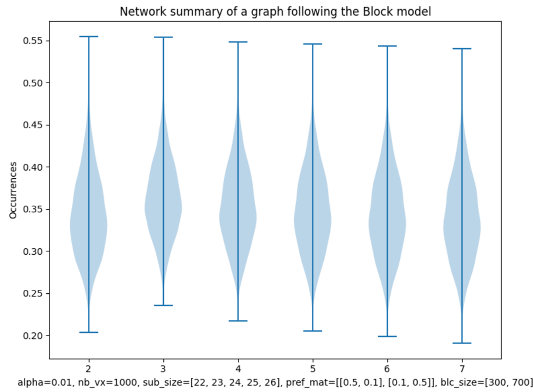

Figure 3: Network summary plot of a graph following the block model, with 1000 nodes, 300 and 700 size of communities, and 0.5 for the edge probability within the community and 0.1 for the edge probability between the communities.

Figure 3 is the network summary plot of a graph following the block model. The particularity of block model graphs is that the number of triangle is greater compared to the one of Erdős–Rényi model graph. Also, other type of cycles is greater as well, but the rate of increase in the occurrence is in the decreasing order with respect to the scale of the counting cycle.

4.1.3 Graphon

The plots of the graphs vary with respect to the choice of graphons.

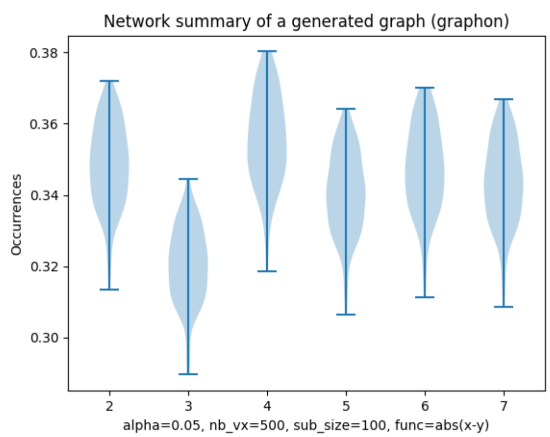

Figure 4: The network summary plot of a graph generated by a graphon.

Figure 4 is the network summary plot of a graph generated by a graphon. In this case the graphon was defined as $W(\xi_i, \xi_j) = |\xi_i - \xi_j|$. The resulting graph had a unique behaviour: the prevalence of triangles was relatively low to the ones of other types of cycles. This phenomenon only happened with this graphon; for more plots with different examples of graphons, see B.3.

4.1.4 Preferential attachment model

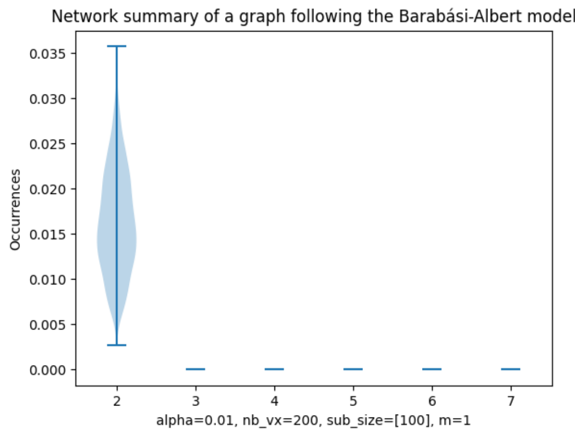

Figure 5: The network summary plot of a Barabási–Albert graph with outgoing edge number equal to one.

Figure 5 is the network summary plot of a preferential attachment model graph with one outgoing edge. Having one outgoing edge would mean that for each new node that we want to consider, it will be only attached to one of the existing node, so the cycles will not be present as it will finally be a tree shape graph. Therefore, we only have occurrences of trees without having cycles. Increasing the number of outgoing edges will then create cycles, and as the number increases, the number of occurrences for each type of tree and cycles increases as well. Also, the occurrences will be in the decreasing order with respect to the number of nodes of the counting cycle. (See Appendix B.4)

4.1.5 Watts-Strogatz model

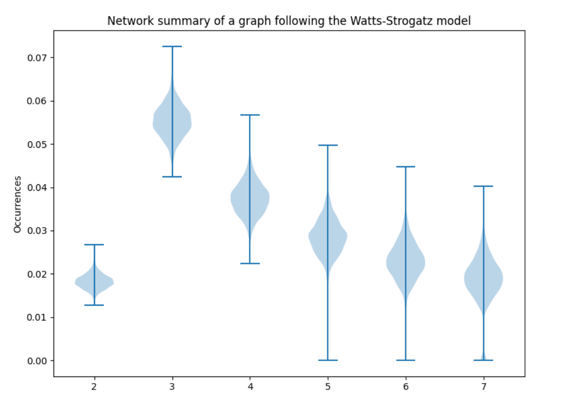

Figure 6: The network summary plot of a graph following the Watts-Strogatz model.

Figure 6 is the network summary plot of a graph following the Watts-Strogatz model. The plot of this model has a particular shape, that the triangle and the rest of the counting cycles have a distinctively higher occurrences compared to the prevalence of trees. It is also possible to see that the tendency is similar to the one of the block model, but in a more extreme behaviour. Furthermore, Increasing the number of neighbors of the nodes will result in the increase of prevalence of trees and cycles. (See Appendix B.5)

4.1.6 Triadic closure

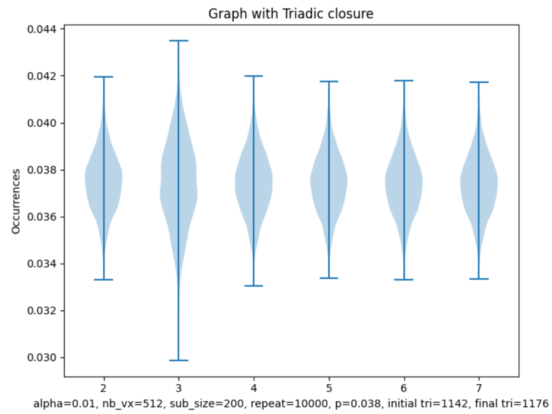

Figure 7: The network summary plot of a graph with the triadic closure.

Figure 7 is a graph with the process of showing triadic closure applied. The applied procedure is the following : starting from an Erdős–Rényi graph of edge probability 0.038, we select uniformly at random 3 nodes, and if only two edges exist between the three nodes, we add the last edge and delete another random edge from the graph to maintain the total number of edges. The graph that was used had this procedure repeated 10,000 times, and by the end the total number of triangles increased as shown on the figure.

4.1.7 Bipartite graphs

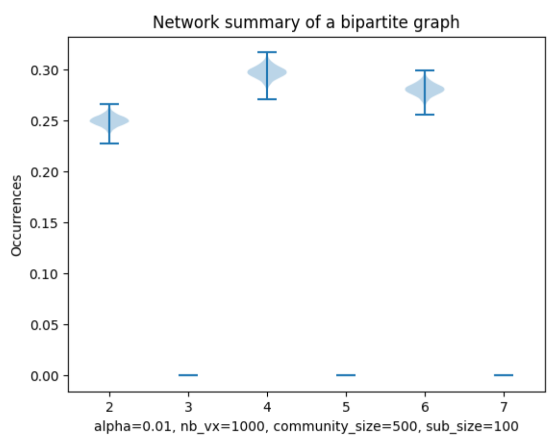

Figure 8: The network summary plot of a bipartite graph.

Bipartite graphs have no triangles or odd-numbered cycles: it is because the nodes within the same community are not connected to each other. Therefore it is necessary to take an even-numbered step for a closed walk. As an effect, the plots only appear on the case of counting the prevalence of trees, and even-numbered cycles, as shown on Figure 8.

4.2 Real data

Data as a form of networks exist in real life, and the network summary plots of those data can give an overview on what kind of model the dataset is following.

Dataset	Nodes	Edges	Edge density	Avg degree
C.elegans	453	4,065	$1.98 \cdot 10^{-2}$	17.9
Power grid	4,941	13,188	$5.40 \cdot 10^{-4}$	5.3
Weblogs	1490	19,025	$1.51 \cdot 10^{-2}$	44.9
Voles	1000-2000	4000-5000	$3 \cdot 10^{-3}$ - $5 \cdot 10^{-3}$	6
Mouse gene	45102	14.5M	$1.56 \cdot 10^{-2}$	670
Retina	1123	577.4K	$9.98 \cdot 10^{-1}$	1.1K
Pollinator	1884	15,265	$8.60 \cdot 10^{-3}$	16.2

Table 4.2 is the table of different real data that have been used with some information on the network.

4.2.1 The metabolic network of C.elegans

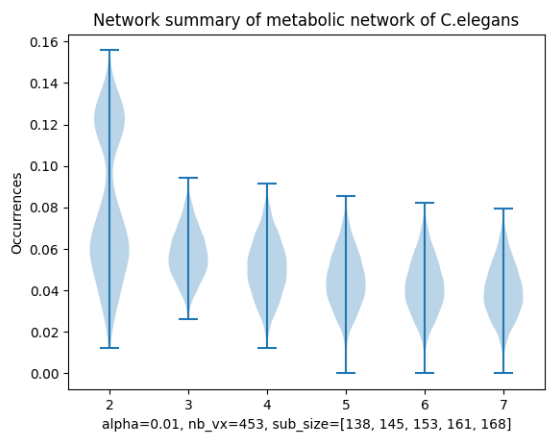

Figure 9: The network summary plot of the metabolic network of C.elegans.

The first data is about the metabolic network of C.elegans[9]. It is shown that the evolution of metabolic network follows the preferential attachment model: newly formed nodes attach more to preexisting well attached nodes[10]. Figure 9 indeed shows that the network resembles to the one of the Barabási–Albert model, with more outgoing edges (See B.4).

4.2.2 The power grid network

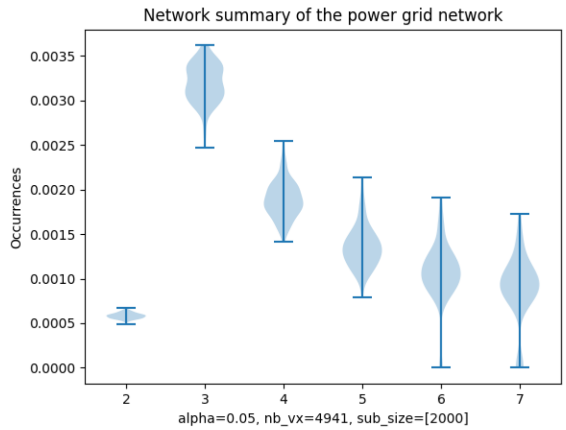

Figure 10: The network summary plot of the power grid network.

The second data is about the power grid network at the western states of the United States [7]. As it is more efficient to connect the different houses with the shortest path lengths from the power source, it is expected to follow the Watts-Strogatz model. Figure 10 shows that first, the graph is very sparse as the occurrences of all trees/cycles are very low compared to usual graphs. As the graph was very sparse, the subsample size was set to 2000 without using the algorithm to determine the sizes. Second, the network is indeed similar to the Watts-Strogatz model.

4.2.3 The network of political weblogs

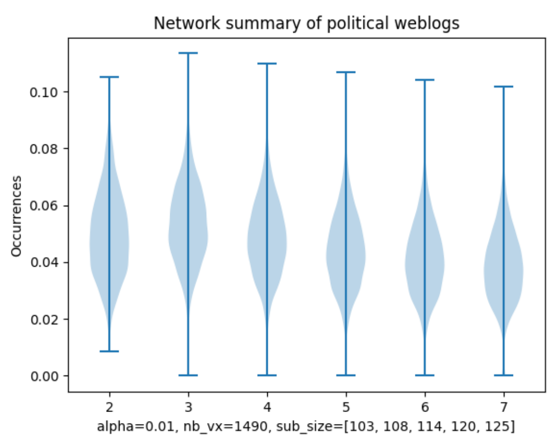

Figure 11: The network summary plot of political weblogs.

The third data is the network of linkage between political weblogs[11]. As the political tendency of people usually lies from conservative to liberal, it is expected to have two communities that would be linked to the one of same political tendency. Figure 11 shows that the data is indeed following the block model.

4.2.4 The network of voles

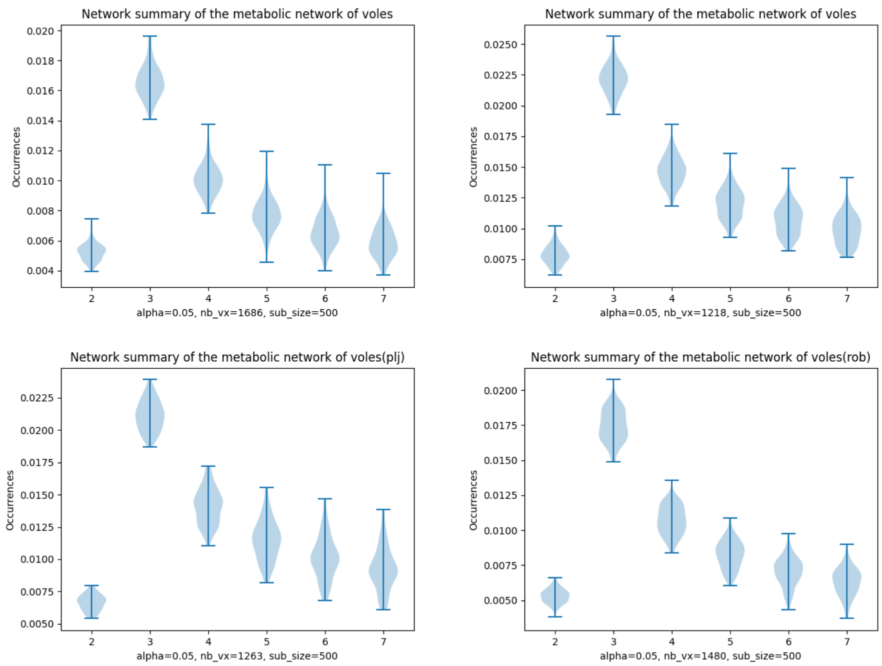

Figure 12: The network summary plot of voles in four different areas.

The fourth data is about the network between voles (a species of rat)[12]. The data was collected in four different areas, considering that two voles are connected when they have been trapped consecutively in the same spot. Although the datasets are from four separate locations, Figure 12 shows that the summary plots all resemble to the one of small world networks, and the prevalence of each trees and cycles are similar as well.

4.2.5 The network of a mouse gene

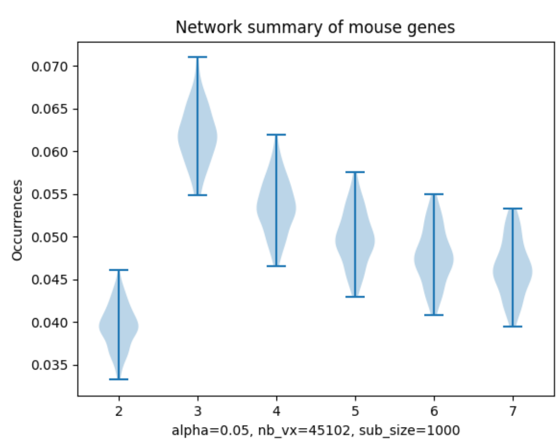

Figure 13: The network summary plot of a mouse gene. The plot is similar to the one of the small world graphs.

The fifth data is about the network of a mouse regulatory gene[13]. The regulatory genes are the genes that regulates the transcription from DNA to RNA. As it controls by the end which proteins to produce from the DNA, it one of the most important part of the genes. Recent studies show that regulatory gene network follows the small-world model[14]. The transcription units have to have a balanced correlation between the units closed to them and the units that are further away which leads the units to form a small-world network. Figure 13 indeed shows that the network follows the Watts-Strogatz model.

4.2.6 The network of a retina

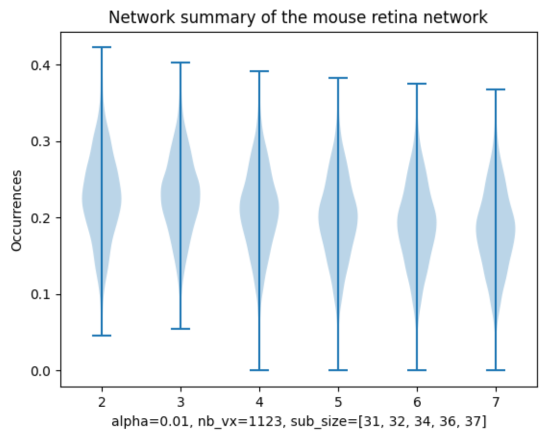

Figure 14: The network summary plot of a mouse retina network. The plot is similar to the one of the small world graphs.

The sixth data is about the mouse retina network[15]. The retina neurons have a complex structure, but they form several clusters, or communities which it is expected that they follow the block model[16]. Indeed, Figure 14 shows that the retina network of the mouse follows the block model.

4.2.7 The network of plant pollinators

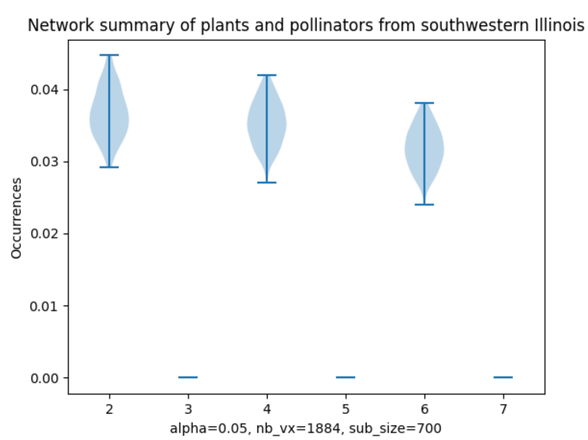

Figure 15: The network summary plot of plant pollinators network. As the graph is bipartite, plots exist only for trees and even numbered cycles.

The last data is about the network between plants and plant pollinators[17]. In the case of plant pollinators, we have two groups : the plants and the pollinators. As we expect the plants to reproduce with the help of the pollinators, and that the pollinators do not pollinate each other, the network of plant pollinators result in a bipartite graph. Figure 15 shows that the graph is indeed bipartite by the absence of odd-numbered cycles.

References

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A Scheme of algorithm 1

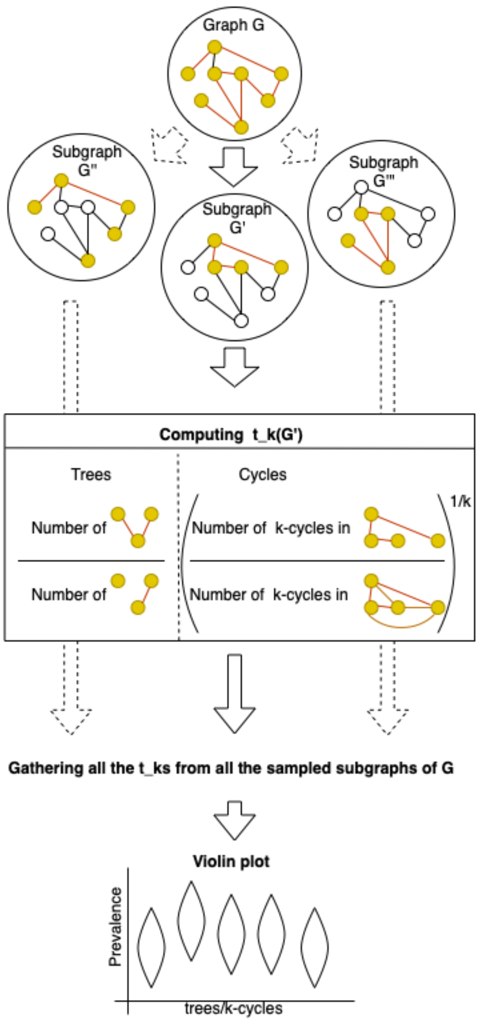

Figure 16: The scheme of algorithm 1. In the case of this scheme, the parameters would be $k_{max} = 4$, $N_s = [4, 5]$. -> appendix?

B Plots of generated models

B.1 Erdős–Rényi model

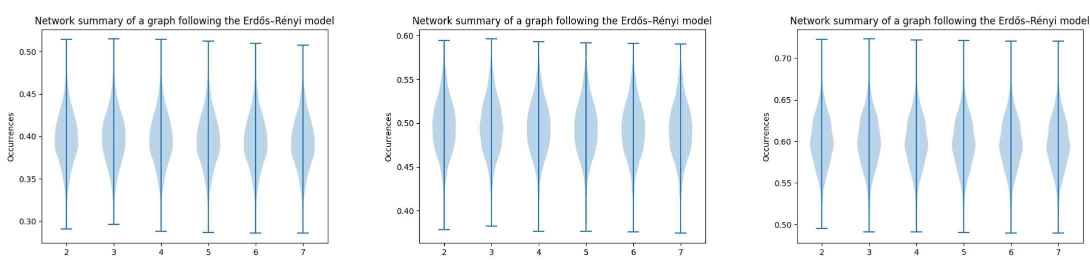

Figure 17: Plots for Erdős–Rényi model graphs with different edge probabilities.

B.2 Block models

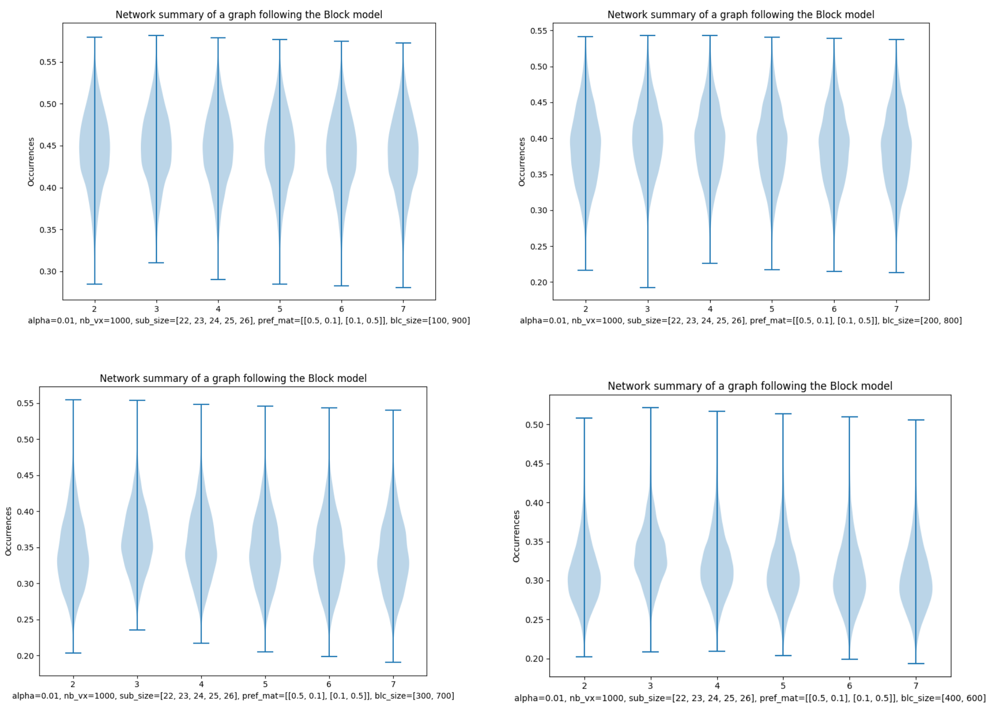

Figure 18: Plots for block model graphs with different size of communities.

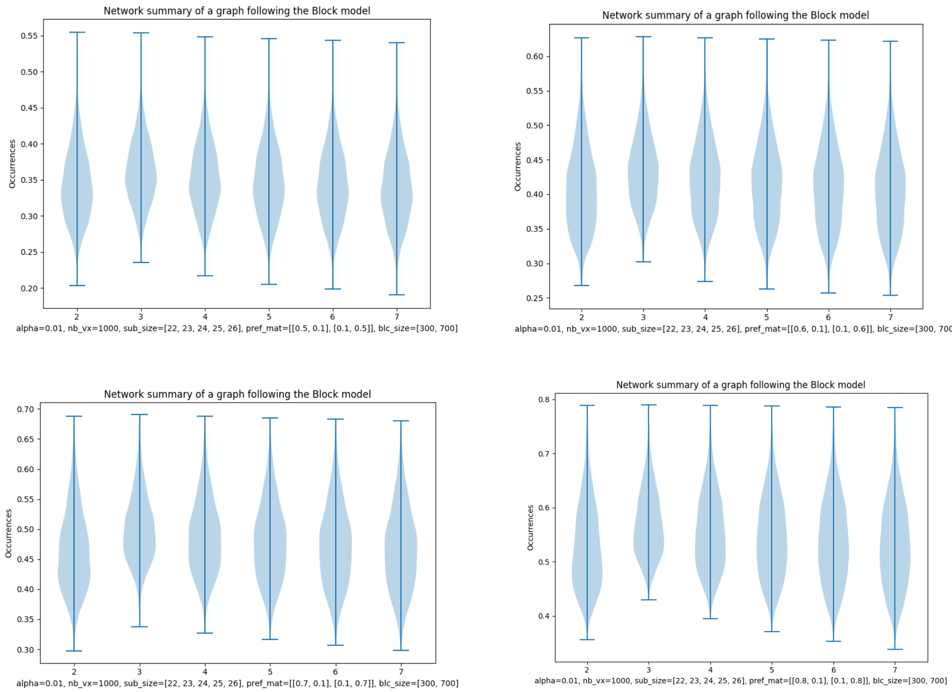

Figure 19: Plots for block model graphs with different probability matrices.

B.3 Graphons

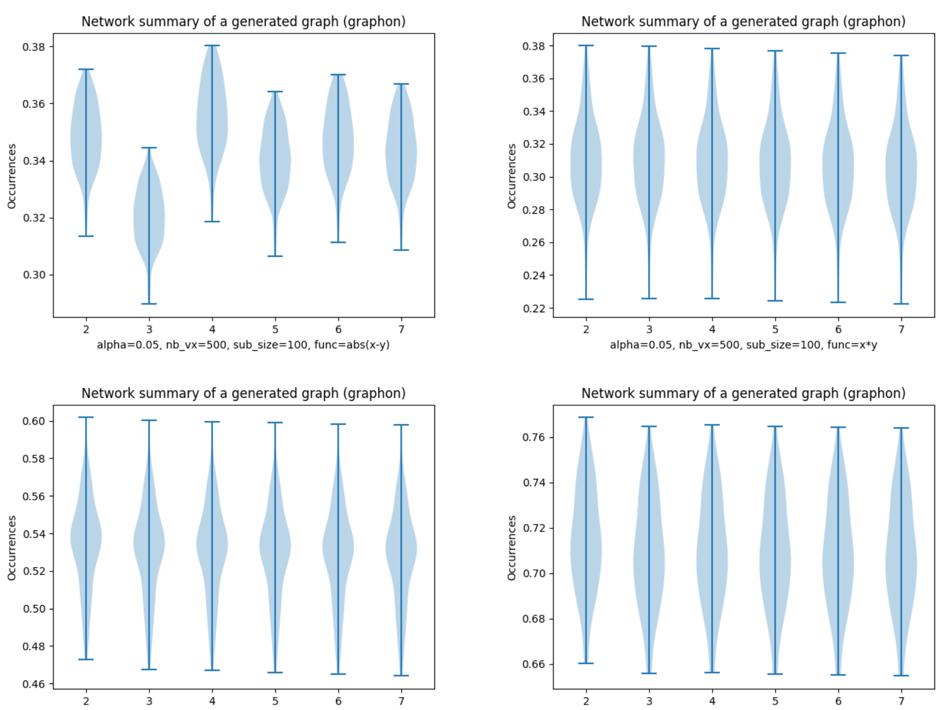

Figure 20: Plots of graphs following a certain graphon function.

B.4 Barabási–Albert models

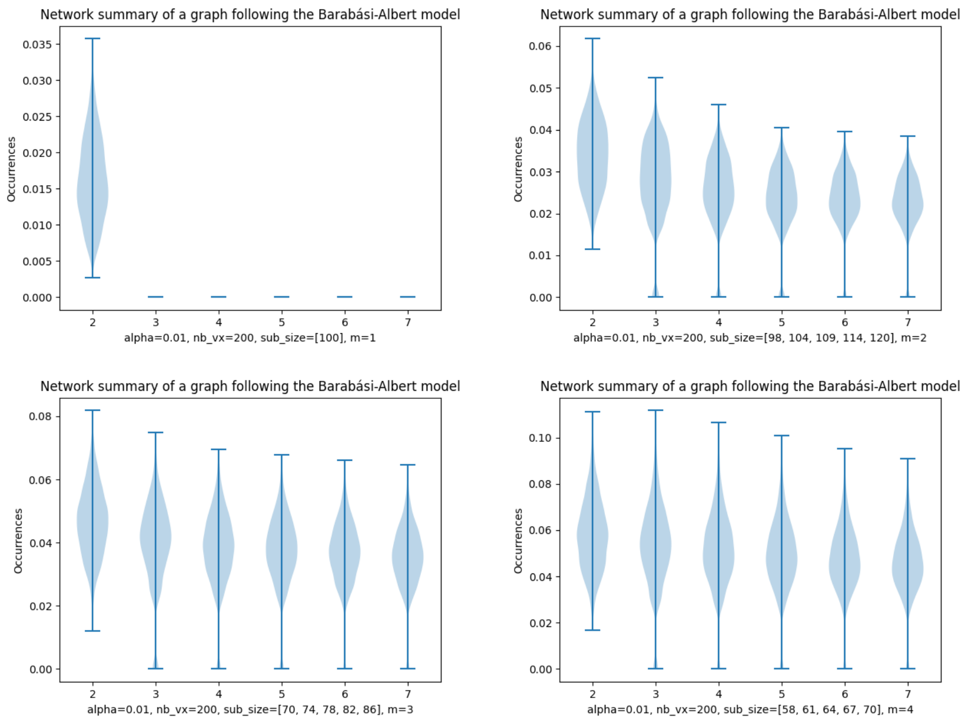

Figure 21: Plots for preferential attachment graphs with different number of outgoing edges.

B.5 Watts-Strogatz models

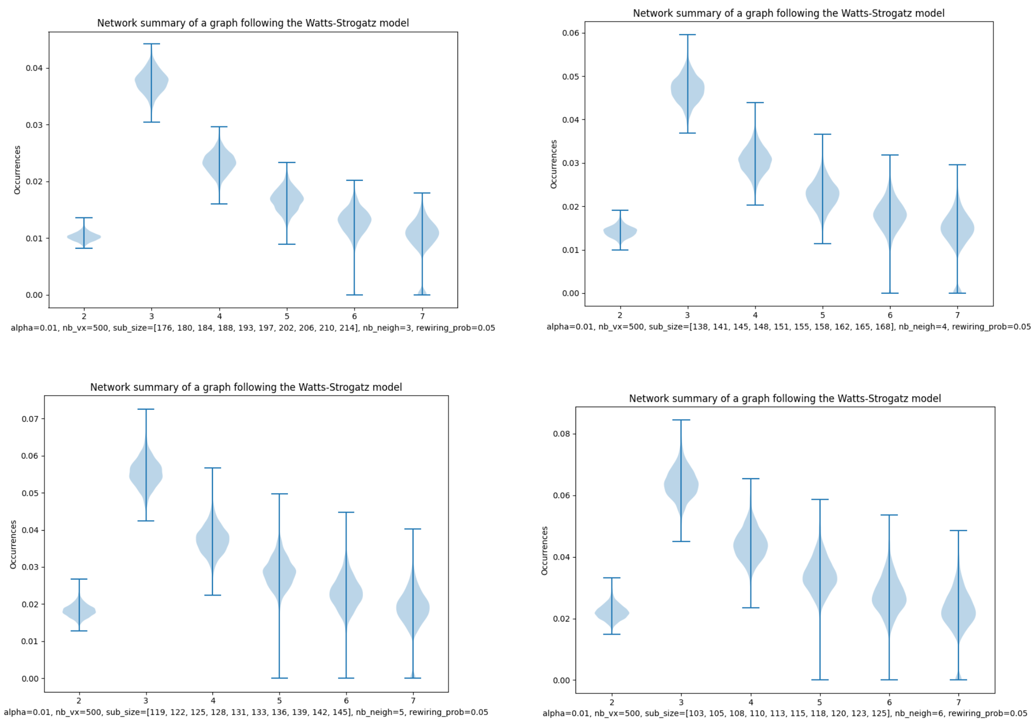

Figure 22: Plots for Watts-Strogatz model graphs with different number of minimum number of vertices to reach all nodes of the graph.

B.6 Bipartite graphs

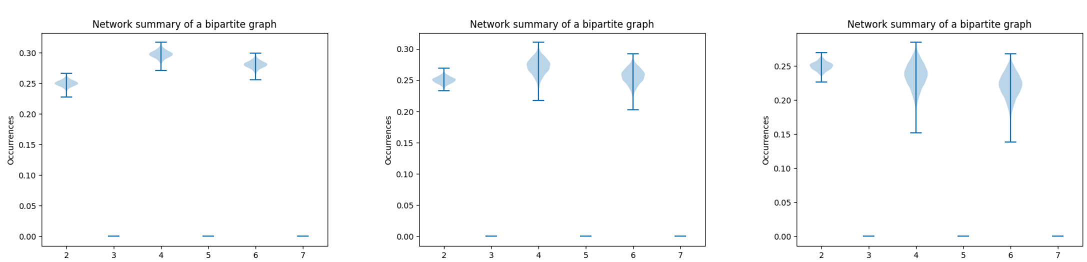

Figure 23: Plots for bipartite graphs with different size of communities.